Backpropagation

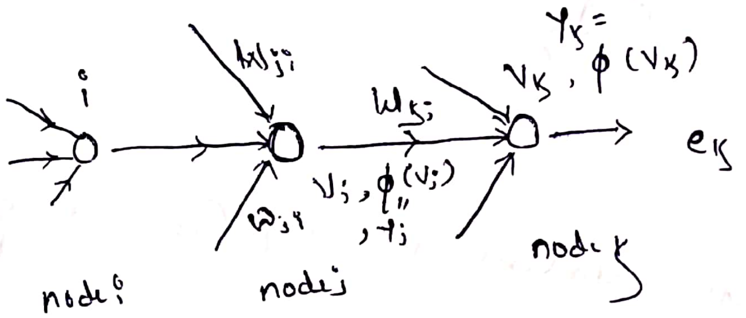
(1)

Backpropagation: "Algorithm that is used to update wt / bias based on Loss value"

Gradient: "rate of change of Error w.r.t wt / bias"

$$\frac{\partial E}{\partial w}$$

Local gradient: "How much given neuron is contributing towards Error"



signal flow diagram

Formula: base

$$e_k^{(n)} = d_{(k)}^{(n)} - y'^{(n)}_{(k)}$$

→ desired, → Expected

$$E(n) = \frac{1}{2} \sum e_k^{2}(n)$$

$$v_k^{(n)} = \sum w_{kj}^{(n)} y_j(n)$$

$$y_k^{(n)} = \phi(v_k(n))$$

output layer → old

$$\Delta w_{kj}^{(n)} = w_{kj}^{(n)} - \alpha \frac{\partial \varepsilon}{\partial w_{kj}} \rightarrow (1)$$

$$\frac{\partial \varepsilon}{\partial w_{kj}} = \frac{\partial \varepsilon}{\partial e_k} \frac{\partial e_k}{\partial y_k} \frac{\partial y_k}{\partial v_k} \frac{\partial v_k}{\partial w_{kj}^{(n)}} \rightarrow (2)$$ {according to chain rule}

$$\frac{\partial \varepsilon}{\partial e_k} = e_k^{(n)} \qquad \frac{\partial e_k}{\partial y_k} = -1 \qquad \frac{\partial y_k}{\partial v_k} = \phi'(v_k(n))$$

$$\frac{\partial v_k}{\partial w_{kj}} = y_j^{(n)}$$

Substitute these values in eq (2) we get

$$\frac{\partial \varepsilon}{\partial w_{kj}} = e_k^{(n)} (-1) \phi'(v_k^{(n)}) y_j^{(n)}$$

$$= -\underbrace{e_k^{(n)} \phi'(v_k^{(n)})}_{\delta_k \rightarrow \text{ local gradient of } k} y_j^{(n)}$$

Sub in Eq (1) we get

$$\boxed{\Delta w_{kj}^{(n)} = w_{kj}^{(n)} + \alpha \delta_k y_j^{(n)}} \rightarrow (3)$$

where $\delta_k = e_k \phi'(v_k(n))$

Hidden layer

11/4

$$\Delta w_{ji}(n) = -\alpha \delta_j y_i(n)$$

$$\delta_j = \frac{\partial \mathcal{E}(n)}{\partial v_j(n)}$$

$$\delta_j = \frac{\partial \mathcal{E}(n)}{\partial e_k} \frac{\partial e_k}{\partial v_k} \frac{\partial v_k}{\partial y_j} \frac{\partial y_j}{\partial v_j}.$$

$$\frac{\delta \mathcal{E}(n)}{\delta e_k} = e_k^n \qquad \delta_j \text{ It } \qquad \frac{\delta e_k}{\delta v_k} = -\phi'(v_k(n)) \qquad e_k(n) = y_k(n) - y'_k(n) = y_k(n) - \phi(v_k)$$

$$\frac{\delta v_k(n)}{\delta y_j} = \sum w_{kj}(n) \qquad \frac{\delta y_j}{\delta v_j} = \phi'(v_j(n))$$

$$\Delta w_j(n) = -\alpha \{ e_k(n) (-\phi'(v_k(n))) \sum w_{kj}(n) \phi'(v_j(n)) \} y_i$$

$$= \alpha \underbrace{e_k(n) \phi'(v_k(n))}_{\delta_k} \sum w_{kj}(n) \phi'(v_j(n)) y_i(n)$$

$$= \alpha \underbrace{\phi'(v_j(n)) \sum_k \delta_k w_{kj}(n)}_{\delta_j} y_j(n)$$

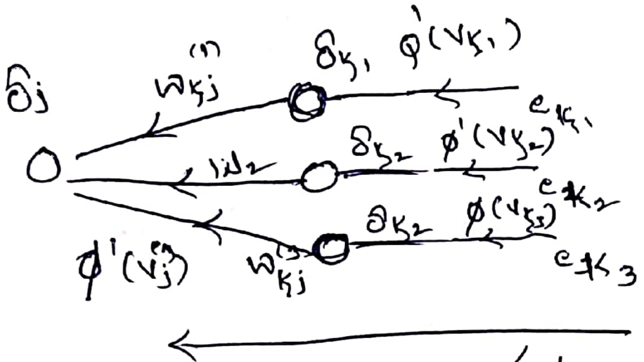
$$\delta_j(n) = \underbrace{\phi'(v_j(n))}_{\text{o/p current node}} \sum_k \underbrace{\delta_k w_{kj}(n)}_{\text{How much Error is coming from its successor}}$$

O/p current node

How much Error is coming from its successor

MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A Constituent unit of MAHE, Manipal)

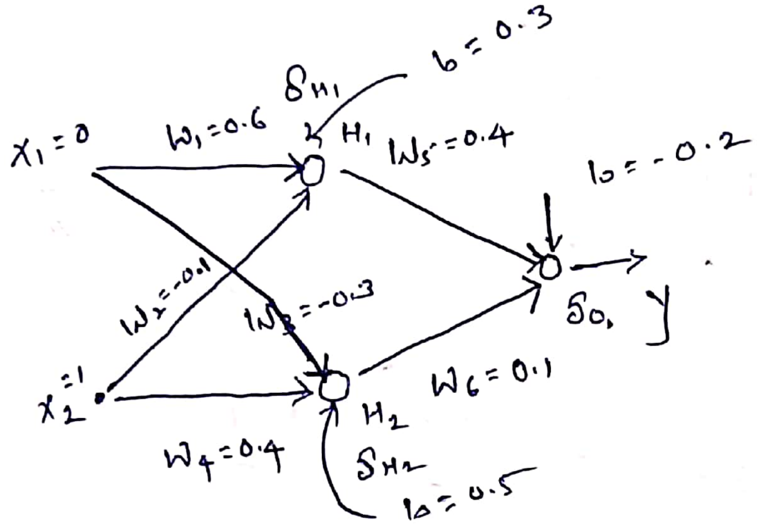
$$\Delta W_{ji}^{(n)} = \alpha \, \delta_j \, y_j^{(n)}$$



Error flow diagram.

Problem :- update wts for one iteration ①

$\alpha = 0.25$ $y = 1$ / sigmoid activation func



Soln Forward pass

a) Hidden layer

$V_{H_1} = 0.3 + (0.6 \times 0 + -0.1 \times 1))$

$= 0.3 + (-0.1))$

$= 0.3 - 0.1 = 0.2$

$\boxed{H_1 = \sigma(0.2) = 0.5498}$

$V_{H_2} = 0.5 + (-0.3 \times 0 + 0.4 \times 1)$

$= 0.9$

$\boxed{H_2 = \sigma(0.9) = 0.7109}$

b) output layer

$V_{O_1} = -0.2 + (0.4 \times 0.5498 + 0.1 \times 0.7109)$

$= 0.09101$

$\boxed{y = \sigma(V_{O_1}) = 0.52277}$

MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A Constituent unit of MAHE, Manipal)
INSPIRED BY LIFE

$$\frac{\partial \sigma(x)}{\partial x} = \sigma x \,(1-\sigma(x))$$

Backward pass

$\Delta W_5 = \alpha \delta_o y_{H_1}$

$= 0.25 \times \delta_o \times 0.5498$

$\delta_o = e_k \phi'(v_k)$ → ~~derivative~~

$= (\cancel{1-0.52277}) \; \cancel{\phi(v_k)(1-\phi(v_k))}$ — $y - y'$

Since activation is sigmoid

$= (1-0.52277)\,\sigma(v_k)(1-\sigma(v_k))$ — $y - y'$

$= (1-0.52277) \; y(1-y)$

$= (1-0.52277) \; 0.52277 \, (1-0.52277)$

$$\boxed{\delta_o = 0.119060}$$

$\Delta W_5 = 0.25 \times 0.119060 \times 0.5498$

$= 0.01636$

$W_5 = 0.4 + 0.01636$

$$\boxed{W_5 = 0.4163}$$

$\Delta W_6 = \alpha \delta_o H_2$

$= 0.25 \times 0.119060 \times 0.7109$

$= 0.02115$

$W_6 = 0.1 + 0.02115$

$$\boxed{W_6 = 0.12115}$$

Hidden nodes

$$\delta_{H_1} = \sigma(V_{H_1})(1-\sigma(V_{H_1})) \times \delta_0 \times 0.4 \qquad \delta_j = \phi'(v_j)\sum_k \delta_k w_{kj}$$

$= 0.5498(1-0.5498) \times 0.119060 \times 0.4$

old wts are used ~~same~~ all since all the wts update same time

$= 0.24751 \times (0.047624)$

$= 0.011781$

llly

$$\delta_{H_2} = \sigma(V_{H_2})(1-\sigma(V_{H_2})) \times \delta_0 \times 0.1$$

$= 0.7109(1-0.7109) \times 0.119060 \times 0.1$

$= 0.0024$

Now

$\Delta W_1 = \alpha\,\delta_{H_1} \times 0 = 0$

$W_1 =$ Same

$W_3 =$ Same

$\Delta W_2 = 0.25 \times \overset{\delta_{H_2}}{\cancel{0.0024}} \times 1$

$= 0.25 \times 0.011781 \times 1$

$= 0.0029$

$W_2 = -0.1 + 0.0029$

$\boxed{W_2 = -0.097}$

$\Delta W_4 = 0.25 \times \delta_{H_2} \times 1$

$= 0.25 \times 0.0024 \times 1$

$= 0.0006$

$\boxed{W_4 = 0.4006}$